

Setup & Configuration

Before starting, configure your identity so your commits are correctly attributed. [\[1, 2\]](#)

- **git config --global user.name "[name]":** Sets the name you want attached to your commit transactions.
- **git config --global user.email "[email address]":** Sets the email you want attached to your commit transactions.
- **git version:** Checks your currently installed Git version. [\[1\]](#)

Starting a Project

Initialize a new project or copy an existing one from a remote server. [\[1, 2\]](#)

- **git init:** Initializes a brand new local repository in your current directory.
- **git clone [url]:** Downloads a project and its entire version history from a remote URL. [\[1, 2\]](#)

The Basic Workflow (Staging & Committing)

This is the daily cycle of saving your work locally. [\[1, 2\]](#)

- **git status:** Lists which files are modified, staged, or untracked.
- **git add [file]:** Adds a specific file to the staging area, preparing it for the next commit.
- **git add .:** Stages **all** modified and new files in the current directory.
- **git commit -m "[message]":** Records your staged changes as a permanent snapshot in history with a descriptive message.
- **git commit --amend:** Modifies the most recent commit (useful for fixing a typo in the message). [\[1, 2, 3, 4\]](#)

Branching & Merging

Branches allow you to work on features in isolation without breaking the main code. [\[1, 2\]](#)

- **git branch:** Lists all local branches in the current repository.
- **git switch [branch-name]:** Switches to the specified branch (modern replacement for git checkout).

- **git switch -c [branch-name]**: Creates a new branch and immediately switches to it.
- **git merge [branch]**: Combines the specified branch's history into your current branch.
- **git branch -d [branch-name]**: Deletes the specified local branch safely. [[1](#), [2](#), [3](#)]

Remote Synchronization

Sync your local work with remote repositories like GitHub or GitLab. [[1](#), [2](#)]

- **git remote add origin [url]**: Connects your local repository to a remote server.
- **git push origin [branch]**: Uploads your local branch commits to the remote repository.
- **git pull**: Fetches changes from the remote and merges them into your current branch.
- **git fetch**: Downloads the latest history from the remote without merging it into your local files. [[1](#), [2](#), [3](#), [4](#), [5](#), [6](#)]

Inspection & Comparisons

Review history and see what has changed over time. [[1](#)]

- **git log**: Displays the full commit history for the current branch.
- **git log --oneline**: Shows a condensed version of the history (one line per commit).
- **git diff**: Shows the differences between files that are modified but not yet staged.
- **git blame [file]**: Shows who last changed each line of a file and when. [[1](#), [2](#), [3](#), [4](#), [5](#)]

Undoing Changes

Use these to fix mistakes or discard unwanted changes. [[1](#), [2](#)]

- **git restore [file]**: Discards unstaged changes in a file, reverting it to the last committed state.
- **git reset --soft HEAD~1**: Undoes the last commit but keeps your changes in the staging area.
- **git reset --hard**: Discards **all** local changes, both staged and unstaged, back to the last commit.

- **git stash:** Temporarily shelves (hides) modified tracked files to clean your working directory.
- **git stash pop:** Restores the most recently stashed changes and removes them from the stash list. [[1](#), [2](#), [3](#), [4](#), [5](#)]